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Instantaneous power

The instantaneous power (P_{inst}) is defined as the power delivered at an instant (as time interval approaches zero),

$$P_{inst} = \frac{dW}{dt}$$

4.3.2 Unit of power

Power is a scalar quantity. Its dimension is $[ML^2T^{-3}]$. The SI unit of power is watt (W), named after the inventor of the steam engine James Watt. *One watt is defined as* the power *when one joule of work is done in one second*, ($1\ W = 1\ J\ s^{-1}$).

The higher units are kilowatt(kW), megawatt(MW), and Gigawatt(GW).

$$1kW = 1000\ W = 10^3 \text{ watt}$$
$$1MW = 10^6 \text{ watt}$$
$$1GW = 10^9 \text{ watt}$$

For motors, engines and some automobiles an old unit of power still commercially in use which is called as the horse-power (hp). We have a conversion for horse-power (hp) into watt (W) which is,

$$1\ hp = 746\ W$$

All electrical goods come with a definite power rating in watt printed on them. A 100 watt bulb consumes 100 joule of electrical energy in one second. The energy measured in joule in terms of power in watt and time in second is written as, 1 J =1 W s. When electrical appliances are put in use for long hours, they consume a large amount of energy. Measuring the electrical energy in a small unit watt. second (W s) leads to handling large numerical values. Hence, electrical energy is measured in the unit called kilowatt hour (kWh).

$$1 \text{ electrical unit} = 1\ kWh = 1 \times (10^3\ W) \times (3600\ s)$$
$$1 \text{ electrical unit} = 3600 \times 10^3\ W\ s$$
$$1 \text{ electrical unit} = 3.6 \times 10^6\ J$$
$$1\ kWh = 3.6 \times 10^6\ J$$

Electricity bills are generated in units of kWh for electrical energy consumption. 1 unit of electrical energy is 1 kWh. (**Note**: kWh is unit of energy and not of power.)

EXAMPLE 4.18

Calculate the energy consumed in electrical units when a 75 W fan is used for 8 hours daily for one month (30 days).

Solution

Power, P = 75 W

Time of usage, t = 8 hour × 30 days = 240 hours

Electrical energy consumed is the product of power and time of usage.